

Object detection model

RAGBOOMS

Overview

- Interactive Flow Chart or Diagram: You could use Tableau's shapes to create a simplified model architecture view (YOLO, Faster R-CNN, etc.), but that's more manual work.
- Alternatively, you can show the performance metrics visually (e.g., accuracy, precision, recall) through bar charts or line graphs.



Data Collection & Preprocessing

- Table or Data Grid:
Show a summary of the dataset (number of images, labels, objects in each class).

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- Bar Chart: Display the number of samples for each class (e.g., how many images contain cars, trucks, pedestrians, etc.).

Real-Time Traffic Insights

Heat Map of Traffic Flow:

- Display a heat map showing congestion based on detected vehicles. This could be done using geographic data (lat/lon of cameras or GPS locations).
- Use color intensity to represent areas with high traffic density.

Time Series Graphs (Traffic Over Time):

- Create line charts to visualize traffic trends (e.g., vehicle counts per hour) to identify peak times of the day, or areas with the most traffic.

Interactive Map with Filters:

- A geographic map of the city, where users can filter by time of day, or specific streets/intersections. This shows traffic detection counts or congestion levels.

Application in Traffic Management

Dashboard with Multiple Charts:

- A combination of bar charts showing detection counts by type (vehicles, pedestrians, traffic signs).
- A line chart for vehicle count over time or by location.
- Dynamic Map with Real-Time Detection: Show a live map where vehicles and objects are dynamically detected and tracked.

Traffic Heatmap (Location-Based Analysis):

- A heatmap that shows areas of highest congestion, allowing stakeholders to focus efforts on busy streets or intersections.

Future work & challenges

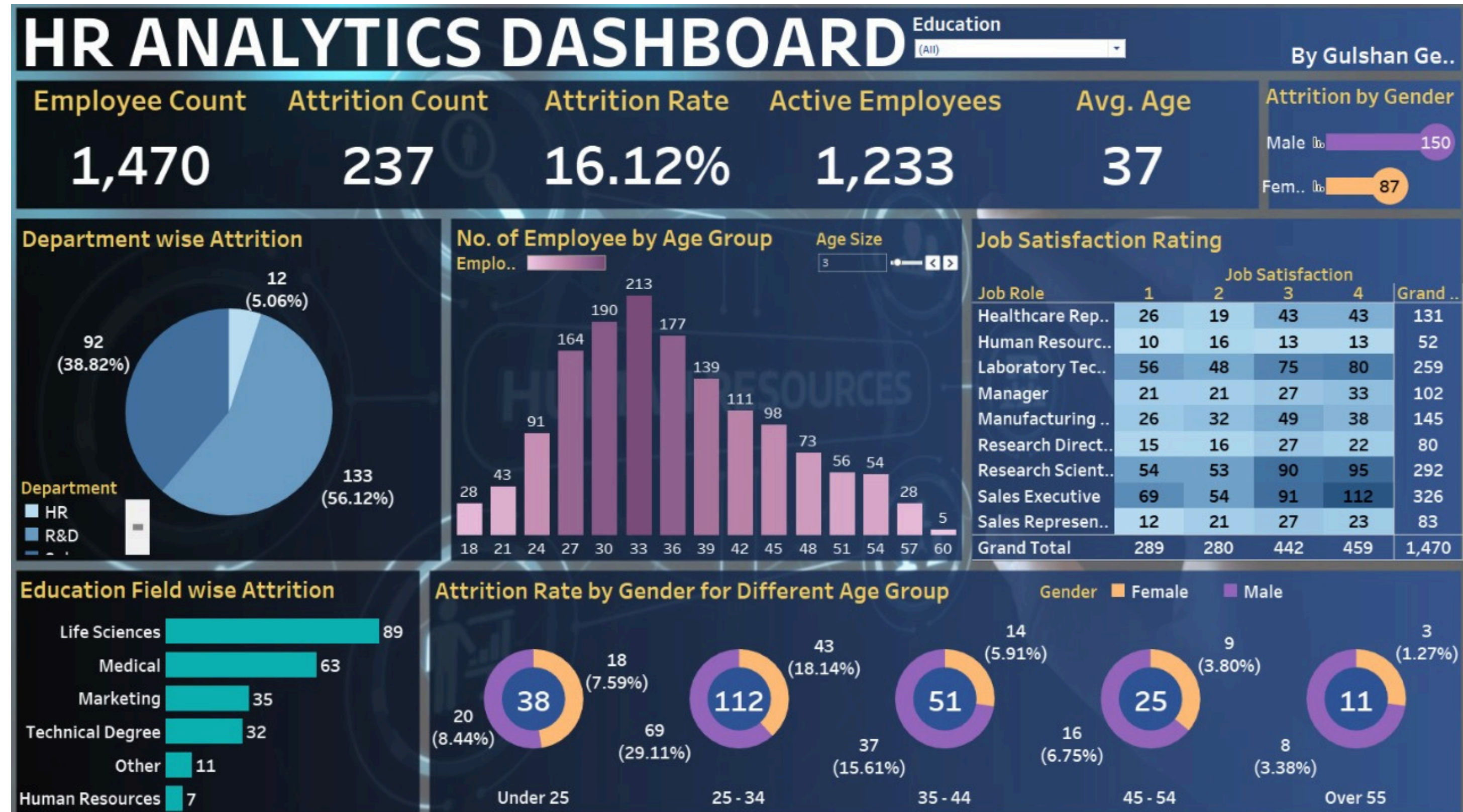
- ◆ **Text Overlay or Annotations:**

Add some annotations or text boxes on the Tableau dashboards highlighting challenges like occlusion, low-light conditions, etc.

- ◆ **Example Graph (Improvement Trends):**

A line graph showing the anticipated improvement in detection accuracy over time, if more data is gathered or the model is fine-tuned.

Tabluae Dashboard



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THANK
YOU